(14) text classification using 1D-CNN with

D1 : Good Movie — (+)

D2 : Good Acting — (+)

D3 : Bad Movie — (−)

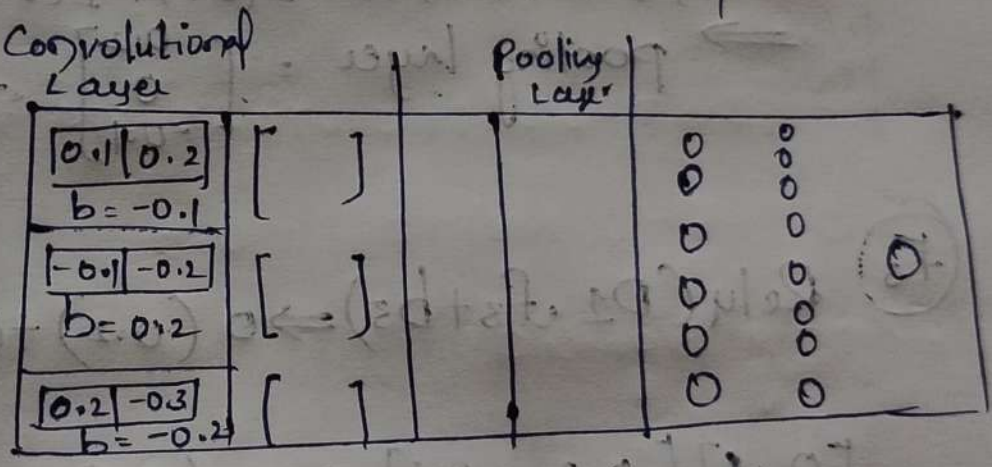
D4 : Bad Acting — (−)

step-1 : BOW

⟹

Doc	Acting	Bad	Good	Movie	label
D1	0	0	1	1	1
D2	1	0	1	0	1
D3	0	1	0	1	0
D4	1	1	0	0	0

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \rightarrow$$



No. of filters = 3

R s = 2

A.F = Relu

stride = 1

$\Rightarrow$ pooling Layer $\Rightarrow \begin{bmatrix} 0.2 \\ -0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}$

Inputs for FNN

$$x = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.2 \\ 0 \\ 0.2 \\ 0.1 \end{bmatrix}$$

Let

$$w_{iH} = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 & 0.1 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.3 & 0.3 & 0.3 & 0.3 & 0.3 & 0.3 & 0.3 & 0.3 \\ 0.4 & 0.4 & 0.4 & 0.4 & 0.4 & 0.4 & 0.4 & 0.4 \\ 0.5 & 0.5 & 0.5 & 0.5 & 0.5 & 0.5 & 0.5 & 0.5 \\ 0.6 & 0.6 & 0.6 & 0.6 & 0.6 & 0.6 & 0.6 & 0.6 \end{bmatrix}$$

$$w_{HO} = \begin{bmatrix} 0.1 \\ -0.1 \\ 0.2 \\ -0.2 \\ 0.3 \\ -0.3 \\ 0.4 \\ 0.4 \end{bmatrix} \qquad b_H = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \\ 0.4 \\ 0.5 \\ 0.6 \\ 0.7 \\ 0.8 \end{bmatrix}$$

(f1)

$D_1 : \Rightarrow Relu(D_1 . f_1 + b_1)$ $\boxed{b = -0.1}$

$\rightarrow 0(0.1) + 0(0.2) - 0.1 = -0.1$

$[0,1][0.1, 0.2] \Rightarrow 0(0.1) + 1(0.2) - 0.1 = 0.1$

$[1,1][0.1, 0.2] \Rightarrow 1(0.1) + 1(0.2) - 0.1 = 0.2$

$\rightarrow$ pooling Layer $= \begin{bmatrix} -0.1 \\ 0.1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$

(f2) $Relu[D_1 . f_2 + b_2] \Rightarrow 0(-0.1) + 0(0.2) + 0.2 = 0.2$

$[0,0][-0.1, 0.2] \Rightarrow 0(-0.1) + 0(-0.2) + 0.2 = 0.2$

$[0,1][-0.1, 0.2] \Rightarrow 0(-0.1) + 1(0.2) + 0.2 = 0.$

$[1,1][-0.1, 0.2] \Rightarrow 1(0.1) + 1(0.2) + 0.2 = -0.1$

$\Rightarrow$ pooling layer $= \begin{bmatrix} 0.2 \\ 0 \\ -0.1 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0 \end{bmatrix}$

(f3) $Relu(D_1 . f_3 + b_3) \Rightarrow 0(0.2) + 0(-0.3) + 0.2$

$= 0.2$

$[0,1][0.2, -0.3] \Rightarrow 0(0.2) + 1(-0.3) + 0.2$

$= -0.1$

$[1,1][0.2, -0.3] \Rightarrow 1(0.2) + 1(-0.3) + 0.2$

$= 0.1$

$Net_H = w_{IH}^{T} \times b_H$

$$= \begin{bmatrix} 0.27 \\ 0.27 \\ 0.27 \\ 0.27 \\ 0.27 \\ 0.27 \\ 0.27 \\ 0.27 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \\ 0.4 \\ 0.5 \\ 0.6 \\ 0.7 \\ 0.8 \end{bmatrix} \cdot \begin{bmatrix} 0.37 \\ 0.47 \\ 0.57 \\ 0.67 \\ 0.77 \\ 0.87 \\ 0.97 \\ 1.07 \end{bmatrix}$$

$O_H = f(Net_H)$ [Relu Activation]

$O_H = \max(0, net_H)$

$net_O = w_{HO}^{T} \cdot O_H + b_O$

$= [0.87 - 0.037 + 0.074 - 0.074 + 0.111 - 0.111 +$

$0.(48 - 0.148) + b_O = 0.5$

Apply sigmoid

function $= \dfrac{1}{1+e^{-0.5}}$

$= \dfrac{1}{1+0.6} = \dfrac{1}{1.6}$

$= \underline{\underline{0.625}}$